

PCM - HUB

Complete manual

Linguists + Administrators

Parametric Comparison Method · Università di Modena e Reggio Emilia
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Contents

1. What is PCM-Hub	6
The three roles	6
2. Logging in	7
First access	7
Changing the password	7
Forgot the password?	7
Session expired	8
Failed-login limit	8
3. Getting around: sidebar, topbar, breadcrumb	9
Sidebar (left)	9
Topbar (top)	9
Footer	10
4. The Dashboard	11
User dashboard	11
Admin dashboard	11
5. Shared pages	13
Glossary (`/glossary`)	13
Instructions (`/instructions`)	14
How to cite (`/how-to-cite`)	15
My Account (`/me`)	16
6. The Languages page	17
Filters (sticky at the top)	17
The table	18
Notes on the map	18
7. Compiling a language	19
Language header	19
Status banner (below the header)	19
Parameter wizard (the squares)	20
A parameter's block	21
Questions	21
Example clipboard	22
Saving a parameter block	23
Switching parameter without saving	23
Submit / Reopen / workflow	23
What an admin sees in addition	24
8. ADMIN ONLY — Parameters	25
Parameter list	25

Edit parameter	26
Deactivating a parameter	26
Implicational condition (notes)	27
Parameters Graph (`/admin/parameters/graph`)	27
9. ADMIN ONLY — Questions	28
Question list	28
Edit/Add question	28
Destructive changes: the archive	29
10. ADMIN ONLY — Motivations	30
List	30
Edits	30
11. ADMIN ONLY — Taxonomy	31
12. ADMIN ONLY — Accounts	32
Account list	32
Add account	33
Safeguards	33
Reassigning languages	33
13. ADMIN ONLY — Table A	34
Modes	34
Filters	34
Manual selection	34
Exports and analyses	35
14. ADMIN ONLY — Filters (Queries Q1–Q10)	36
Q3 — the richest	37
Q10 — answers and examples	37
15. ADMIN ONLY — History, Backups, Migration, Backup Restore, Import Excel, Recompute	38
History & Backups (`/admin/history`)	38
Migration Import (`/admin/migration-import`) — dangerous	42
Backup Restore (`/admin/backup-restore`)	43
Import from Excel (`/admin/import-excel`)	44
Recompute final values	44
16. Common errors and what to do	45
"Form locked by the current language status"	45
"Question X needs at least 2 valid examples when answering YES or UNSURE"	45
"This block has been modified by another session..."	45
"Wrong email or password"	45
"Too Many Requests" at login	45
"Invalid email format"	45
"You cannot delete your own account"	45

"You cannot delete the last remaining administrator"	45
The "Submit for approval" button is missing	45
The site logged me out unexpectedly	45
I changed the taxonomy but languages still show the old names	46
17. Quick interface glossary	47

Practical guide to using the **PCM-Hub** website (hub.parametriccomparison.unimore.it). This manual is for daily users: linguists who compile their assigned languages, and administrators who manage parameters, questions, accounts and cross-language analyses.

You'll find what you need faster if you know your role:

- **Linguist (User)** → read sections 1-7 and jump straight to §7 (Compiling a language).
 - **Administrator (Admin)** → read everything: admins see all User functionality plus a dozen extra tools (sections 8-15).
-

1. What is PCM-Hub

The **Parametric Comparison Method** (PCM, Longobardi & Guardiano 2009) compares different languages using a set of **universal syntactic parameters**. Every language is “compiled” by a linguist who, for each parameter, answers a set of questions (mostly yes/no) and provides linguistic examples to support the answers.

From the data collected, the website automatically produces:

- consolidated parameter values for each language (+, −, 0, ?);
- comparative matrices across languages;
- Hamming, Jaccard and geographic distances;
- dendrograms and cluster maps;
- PCA analyses, Mantel tests.

This makes it possible to study phylogenetic relationships, divergences and convergences across languages from different families.

The three roles

Role	What they can do
Public (visitor, not logged in)	sees only the home page with the interactive map and the How to cite page
User (assigned linguist)	compiles their assigned languages, accesses glossary, instructions, language list, personal dashboard
Admin	everything a User does, plus: parameters/questions/motivations/taxonomy management, account creation, TableA, queries Q1-Q10, history/backup, migration, recompute

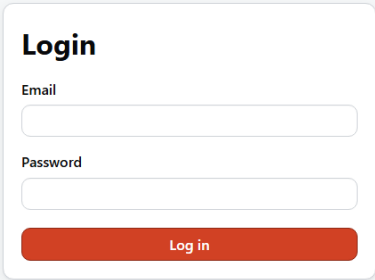
2. Logging in

First access

The administrator who created your account will give you:

- the **email** of your account;
- a **temporary password** (at least 8 characters).

Go to the site URL → click **Login** from the home page (or go straight to `/login`) → enter the credentials → **Log in**.

A screenshot of a login form titled "Login". It features two input fields: "Email" and "Password". Below the "Password" field is a red button labeled "Log in". The form is centered on a light gray background.

On your first login **we recommend changing your password**: top-right click **MyAccount** → **Change password**.

Changing the password

From the **My Account** page you have two blocks:

- **Profile**: name, surname, email (if you change the email it becomes the new login credential).
- **Change password**: requires the current password + the new one twice. Minimum 8 characters.

Forgot the password?

For now **email reset is not yet active** (it requires the SMTP integration from Unimore IT services). In the meantime: contact an administrator, who can reassign you a temporary password by creating a new account or intervening from the panel.

Session expired

The access token lasts **30 minutes of inactivity**. After that, protected pages will automatically send you back to the login. Nothing is broken, just log in again.

Failed-login limit

For security: **after 5 failed login attempts from the same connection within 1 minute**, the system replies with a “Too Many Requests” error for a few seconds. Wait a moment and retry.

3. Getting around: sidebar, topbar, breadcrumb

Once logged in, the interface has 3 fixed elements:

Sidebar (left)

All the main pages. The visible items depend on your role:

Item	User	Admin
Dashboard	✓	✓
Citation Guidelines	✓	✓
Languages	✓	✓
Parameters	—	✓
Parameters Graph	—	✓
Questions	—	✓
Motivations	—	✓
Taxonomy	—	✓
Table A	—	✓
Filters	—	✓
Accounts	—	✓
History & Backups	—	✓
Migration Import	—	✓ (red = dangerous)
Backup Restore	—	✓
Instructions	✓	✓
Glossary	✓	✓

The **button at the top of the sidebar** collapses/expands it (useful to gain horizontal space, especially in TableA). The preference is remembered.

Topbar (top)

- **Breadcrumb** on the left: shows the page path (e.g. `Dashboard / Languages / Italian / Data`). Items in **blue** are clickable; items in **grey** are informational only (e.g. dynamic IDs).
- **MyAccount** on the right: shortcut to the profile page.
- **Theme toggle (Light/Dark)**: switches the visual theme. The preference is remembered too.
- **Logout**: ends the session and brings you back to the public home.

Footer

At the bottom of every page:

- PCM-Hub citation (CC BY 4.0);
 - links to Privacy Policy, Disclaimer, lab contact email (`pcm_lab@unimore.it`);
 - link to the institutional PCM Unimore project website;
 - **Role badge** (bottom right): tells you whether you are “Admin Access”, “User Access” or “Public View”.
-

4. The Dashboard

It's the first page you see after login.

User dashboard

Shows **the languages assigned to you**. For each language:

- ID and full name (e.g. **ITA – Italian**);
- **status** (**Pending** , **Waiting** , **Approved** , **Rejected**);
- **progress bar** (number of answers given / total questions, percentage);
- **Fill in** button (or **View rejection and reopen** if the language was rejected) leading to the compilation page.

If the language is **Rejected** , below you'll also see the **admin's note** explaining why it was rejected.

Admin dashboard

The PCM Hub Dashboard MyAccount Dark Logout

Welcome, Admin

WAITING FOR APPROVAL
0
No waiting reviews.

LANGUAGES BY STATUS

111 PENDING 0 WAITING 3 APPROVED 0 REJECTED
Total: 114

FLAGGED/UNSURE PARAMETERS
Unsure or partially compiled (empty parameters excluded).
245

Russian	4
Hungarian	2
Finnish	2
Kazakh	3
Basque Central	1

LATEST CHANGES

Param	Note	By	Date
FGM	modificato	Admin	06/05/2026
FGP	[Question FGP_Qg - Motivation MOT014] Global change: description updated	Admin	06/05/2026
FGM	[Question FGM_Qb] aggiunta motivations	Admin	02/05/2026
FGK	[Question FGK_Qa - Motivation MOT002] Global change: description updated	Admin	02/05/2026
FGA	[Question FGA_Qa - Motivation MOT002] Global change: description updated	Admin	02/05/2026
FGA	[Question FGA_QSa - Motivation MOT002] Global change: description updated	Admin	02/05/2026
FGA	[Question FGA_Qa] modifica qu	Admin	01/05/2026
FGK	[Question FGK_Qa - Motivation MOT002] Global change: description updated	Admin	01/05/2026
FGA	[Question FGA_QSa - Motivation	Admin	01/05/2026

3+1 card layout:

- **Waiting for Approval:** count of languages awaiting review + quick links to open them.
- **Languages by Status:** clickable counters **Pending** / **Waiting** / **Approved** / **Rejected** . Clicking a counter expands a panel listing the languages in that status.
- **Flagged/Unsure Parameters:** count of “red” parameters (partial compilation or marked Unsure) grouped by language, with totals.
- **Latest Changes** (right): table with the most recent parameter changes (who, when, what, note).

From here an admin can quickly open languages that need attention.

5. Shared pages

Glossary (/glossary)

The PCM Hub

Dashboard
How to cite (edit)

DATA & TOOLS

Languages
Parameters
Parameters Graph
Questions
Motivations
Taxonomy
Table A
Filters
Accounts
History & Backups
Migration Import
Backup Restore
Instructions
Glossary

Dashboard / Glossary

MyAccount Dark Logout

Glossary

Search by term or description... Add Term

Term	Description	Actions
(morphological) Case	In nominal morphology it is important to have a demarcation criterion between inflectional (lato sensu, i.e. including agglutinative morphology) Case proper and pre-/post-positional words. Two criteria...	Edit Delete
Article	It is used to refer to a determiner that does not express any meaning other than (in)definiteness or just phi-features, and sometimes even less interpretable content (expletive articles). In some languages, articles ...	Edit Delete
Atomizing	The process shifting the interpretation of a nominal argument from an unbounded reading (a free variable, indirectly bound) to a bounded one (a variable bound by a specific determiner).	Edit Delete
Bare noun	A noun (or its extended maximal phrase) not introduced by any overt determiner in the canonical Determiner position nor with a determiner-like (atomizing) interpretation derived at a distance from other elemen...	Edit Delete
Bounded vs. unbounded reading	A nominal argument will be said to have an unbounded reading when it is at the same time obligatorily indefinite, scopeless, and atelic. This reading is typically instantiated in European languages by bare plural ...	Edit Delete
Case suffixes vs. postpositions	How to distinguish case suffixes from postpositions in postpositional languages: \n1. the case suffix occurs on a head noun or adjective before some other phrase-internal word (another adjective, a relative ...	Edit Delete

Alphabetical list of technical terms used in the project, with descriptions.

- All users can **search** (search field at the top).
- Admin only: **Add term**, **Edit** and **Delete** buttons on rows.

Instructions (/instructions)

The screenshot displays the 'Instructions' page within the PCM Hub interface. The sidebar on the left contains navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, **Instructions**, and Glossary. The main content area is titled 'Instructions' and includes an 'Edit' button. The content is organized into sections: 'General instructions for entering data' and 'Specific instructions for writing glosses'. The 'General instructions' section contains five numbered steps: 1. Read the question carefully. 2. Identify the corresponding structure in your language. 3. If your language: (a) has examples, select YES and provide at least two examples; (b) has no examples, select NO and provide an ungrammatical example or motivation. 4. After completing each parameter, save your answer and data by selecting a confidence button (green for complete, red for unsure). 5. Click Submit only after all data have been entered. The 'Specific instructions for writing glosses' section advises focusing on relevant words and using provided glosses, with a link to 'Leipzig Glossing Rules'. At the bottom, a table with columns 'Gloss' and 'Meaning' is partially visible.

Editable HTML document (powered by TinyMCE) with the compilation guidelines. The default version includes:

- general rules (read → identify → answer YES/NO → save → submit);
- specific instructions for **linguistic glosses** with a symbol table (ACC, NOM, GEN, etc.);
- glossing examples.

Admin only: **Edit** button at top right → WYSIWYG editor with formatting, tables, raw HTML. **Save** persists the document for everyone.

How to cite (/how-to-cite)

The screenshot shows the PCM Hub interface. The top navigation bar includes 'The PCM Hub' logo, a breadcrumb 'Dashboard / How to cite', and user options 'MyAccount', 'Dark', and 'Logout'. The left sidebar contains a 'Dashboard' button and a 'How to cite (edit)' button. Below these are 'DATA & TOOLS' links: Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, and Glossary. The main content area is titled 'Citation Guidelines' and features two cards. The first card, 'Parameters & Manifestations', is for Admins and Users and provides a citation example for a journal article with a 'Copy' button. The second card, 'Updated language list & map, analyses, graphs and scripts', is for Public and Admins and provides a citation example for the PCM Hub itself with a 'Copy' button. Both cards have an 'Edit reference' link at the bottom.

Two cards:

- **Parameters & Manifestations:** how to cite the PCM Hub parameters/questions (recommended for User and Admin).
- **Database** (linguistic data): how to cite the linguistic data collected.

Below each card there is a **Copy** button that copies the full citation to the system clipboard.

Admin only: **Edit** button opens the TinyMCE editor for the citation text.

My Account (/me)

The screenshot shows the 'My Account' page in the PCM Hub application. The sidebar on the left contains the following links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import (highlighted in red), Backup Restore, Instructions, and Glossary. The main content area has a breadcrumb 'Dashboard / My Account' and a user menu with 'MyAccount', 'Dark', and 'Logout' options. The 'My Account' section is divided into two parts. The 'Profile' section contains input fields for 'Name' (with 'Admin' entered), 'Surname', and 'Email' (with 'admin@test.com' entered), and a 'Save Profile' button. The 'Change Password' section contains input fields for 'Current Password', 'New Password (min 8 characters)', and 'Confirm New Password', and an 'Update Password' button.

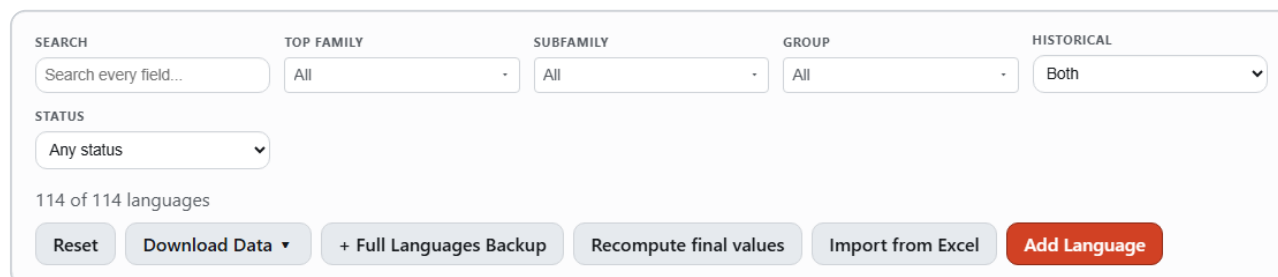
- **Profile:** name, surname, email. **Save** writes directly to your profile.
- **Change password:** old password + new one twice. Minimum 8 characters. If the two new copies don't match, the system rejects.

⚠ If you change the email, the **new email becomes your login credential**. Memorize it well (or use a password manager).

6. The Languages page

The list of all the languages in the project. **Above** the table there is an **interactive map** (OpenStreetMap) with a pin for each georeferenced language.

Filters (sticky at the top)



SEARCH: Search every field...

TOP FAMILY: All

SUBFAMILY: All

GROUP: All

HISTORICAL: Both

STATUS: Any status

114 of 114 languages

Reset Download Data + Full Languages Backup Recompute final values Import from Excel Add Language

- **Search** global text (ID, name, family, top family, group, status, rejection note).
- **Top Family / Subfamily / Group**: hierarchical multi-select. Choosing a top-family narrows the available subfamilies, and so on.
- **Historical**: all / only historical / only non-historical.
- **Status**: any / Pending / Waiting / Approved / Rejected.

To the right of the filters: counter “X of Y languages”, and the buttons:

- **Reset**: clears filters.
- **Clear exclusions** (appears if you’ve manually excluded languages).
- **Download Data** ▾ (admin): opens a menu to export data for the **currently filtered, non-excluded** languages:
 - **Export language metadata (.xlsx)** — ID, name, family, coordinates, etc.
 - **Export backup (.zip)** — full bundle with all answers and examples for the selected languages (async job: a toast at the bottom-right shows progress).
 - **Map (.png)** — the current map as an image.
 - **Geographic distances (.txt)** — GCD km matrix between the selected languages.
- **+ Full Languages Backup** (admin): creates a DB-side snapshot of **all** languages (asks for an optional note). You’ll find it later under History & Backups → Full backups.
- **Recompute final values** (admin): re-runs the implicational DAG for **all** languages. Async job with progress bar. Use after important changes to implicational conditions, or after an import.
- **Import from Excel** (admin): structured import page.
- **Add Language** (admin): create a new language manually.

The table

☒	ID	Name	Status	Top family	Subfamily	Group	Actions			
☒	Afk	Afrikaans	Approved	Indo-European	Germanic	West Germanic	Data	Duplicate	Debug	Edit
☒	Aid	Aidone	Pending	Indo-European	Romance	Gallo-Italic of Sicily	Data	Duplicate	Debug	Edit
☒	Ar	Arabic	Pending	Semitic	West Semitic	—	Data	Duplicate	Debug	Edit
☒	Arc	Archi	Pending	NE-Caucasian	Lezgic	—	Data	Duplicate	Debug	Edit
☒	BA	Bari	Pending	Indo-European	Romance	Apulo-Barese (Upper Southern)	Data	Duplicate	Debug	Edit

- **Left checkbox:** toggle “include / exclude” of the language from the effective set (map, distances, exports). Ticking the header excludes/includes all visible languages.
- **ID, Name, Status, Top family, Subfamily, Group.**
- **Actions:**
 - **Data:** opens the compilation page (regular users and admins).
 - **Duplicate** (admin): duplicates the language with all answers and examples (useful as a template).
 - **Debug** (admin): opens a page showing `init` / `final` values of every parameter for the language, with the option to **Apply implicational condition(s)** (re-runs the DAG on this single language).
 - **Edit** (admin): edits the language metadata (name, coordinates, family, etc.).

Notes on the map

Markers are colored by **top-family**. The legend below the map lists every top-family with its color: hovering a label dims the others (useful to visually isolate a family).

7. Compiling a language

This is the **heart of the site**: where a linguist enters and answers the questions for a language. You get here from:

- Dashboard → click on an assigned language;
- Languages → row → **Data** button.

The URL is `/languages/<id>/data`.

Language header

The screenshot shows the PCM Hub interface. On the left is a sidebar with 'The PCM Hub' logo and a menu including Dashboard, How to cite (edit), DATA & TOOLS (Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, Glossary), and a user profile section. The main content area has a breadcrumb 'Dashboard / Languages / Aid / Data' and user controls 'MyAccount', 'Dark', and 'Logout'. The language header for 'Aidone (Aid)' is displayed, with an 'Export parametric data (.xlsx)' button. Below this is a table of metadata:

TOP-LEVEL FAMILY	Indo-European	LONGITUDE	14.44
SUBFAMILY	Romance	SUPERVISOR	Cristina Guardiano, Vincenzo Stalfieri
HISTORICAL	No	INFORMANT	Franca Ciantia
ISO CODE	scn	GROUP	Gallo-Italic of Sicily
GLOTTOCODE	east2279	SOURCE	—
LOCATION	Aidone	ASSIGNED TO	—
LATITUDE	37.42		

Below the metadata table is a 'Pending' section with the text 'You are filling in this language. Changes persist between sessions.' and buttons for 'Apply implicational condition(s)' and 'Change Status'. At the bottom is a grid of 48 language codes (FGM, FGA, FGK, SPK, FGP, FSP, FGN, SCO, GDP, FSN, FNN, FGT, FGG, FSG, CGB, FPC, DGR, DGP, CGR, NWD, FVP, DGD, DPQ, DCN, DNN, DIN, FGC, FGE, FCN, HMP, ARR, GCN, GFN, GFP, GP3, GEI, CSE, EAL, CAL, LKA, LKO, LKP, DMP, DMG, GUN, GAD, GFL, PGL, GGH, GSI, ALP, GIT, UST, GPC, PSC, PCA, PMN, RHM, FRC, NRC, DOR, FFP, NUP, PNP, NUD, NUC, NM1, EAF, NM2, NUA, NGL, ACM, DSN, DSA, DSS, DOC, NEX, PEX, FEX, PDC, PCL, APO, WAP, AGE, OPK, TSP, TDP, TDC, TSA, TAR, TLC, TND, TDA, TNL, FBN, FVN, FPO, PPK, POK, TPP, LPO, PGE, DPP, GMD, GMC, CEX, PKE, PKP).

At the top you see:

- **Name and ID** of the language;
- **Export parametric data (.xlsx)** button (admin) or **Export examples (.xlsx)** (user) to download an Excel with the data you've entered;
- a grid with the language **metadata** (top-family, family, group, ISO/Glottocode, coordinates, supervisor, informant, source, etc.). Only admins also see the assigned user.

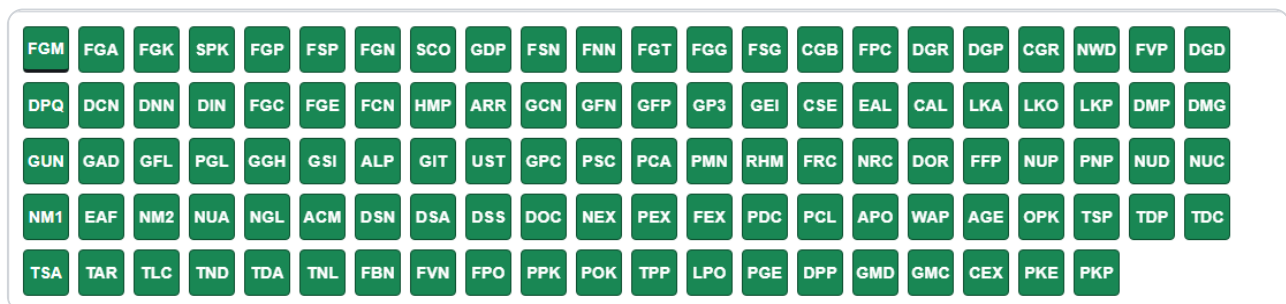
Status banner (below the header)

Shows the current status of the language, a short description and the available workflow buttons. The 4 possible statuses:

Status	Meaning	What you can do (assigned User)
Pending	Being compiled, editable	write/edit freely; when done click Submit for approval
Waiting for approval	Submitted, awaiting admin review	the form is locked , you have to wait. The admin will receive the language in their Dashboard widget
Approved	Approved, frozen	the form is locked read-only. The language joins the official dataset.
Rejected	Rejected, needs revision	read the admin's note, click Reopen to put the language back into Pending and resume editing

⚠ **Admins instead** can always edit regardless of status (they see a purple “Admin override” banner indicating this). Admins also have a **Change Status** ▾ dropdown that lets them force the language to any status (Approve/Reject/Pending/Waiting), bypassing the normal workflow.

Parameter wizard (the squares)



Below the status banner there is a **row of small squares**, one per active parameter. Clicking a square jumps to **that parameter's question block**.

Squares are colored by progress:

- **Grey (empty)** — no answers yet.
- **Yellow (in progress)** — some questions answered but not all, or parameter marked as “Unsure”.
- **Green (complete)** — every question has a valid answer (**yes** or **no** ; **unsure** does not count as completed).

Tip: use the colors as a visual to-do list. All green = ready to submit.

A parameter's block

FGM — grammaticalized morphology

Distinguishes languages that have words containing bound morphemes for grammatical meanings (e.g., IE, Uralic, Semitic, Japanese) from languages that do not (e.g., Mandarin, Cantonese)

ADMIN NOTES

FGM_Qb
 Does the language have roots which take different affixes/phonological alternations encoding different closed-class interpretable/grammatical properties (Tense, Aspect, Number, Gender, Gradation, Case, etc.)?

Instructions
Is there inflectional morphology?

Example YES

1. cat - cats
2. sing - sang

Answer

YES

Examples

Copy Remove
 Example #1
 Example text
 pindza
 Transliteration

Copy Remove
 Example #2
 Example text
 pindza
 Transliteration

When you click a square you see:

- header with **ID and name** of the parameter, plus a `short_description`;
- (admin only) collapsible **Admin notes** area where the admin can write a free-text note for this (language, parameter) pair. The note is not visible to Users and is not exported.
- **one card per question** in the parameter;
- at the bottom-right a **floating ("sticky") toolbar** with two save buttons: **Confident** → **Next** (green) and **Unsure** → **Next** (red). See below.

Questions

Each question shows:

- a header with the **ID and text** of the question;
- a **More info** button (if there is help text, expandable);
- **Instructions**: general instruction, always visible;
- **Example YES**: an illustrative example, always visible (decorative only, you don't fill it);

- **Instructions (YES) or Instructions (NO):** specific instructions, shown only after you select the corresponding answer;
- **Answer** (dropdown): – select –, YES, NO, UNSURE. This is your answer.

Depending on the chosen answer, the card expands differently:

You selected NO

The **Motivations** section appears: a list of checkboxes with predefined reasons why this question is “No” for this language (e.g. “no corresponding structure”, “marginal phenomenon”). Tick one or more. If the question has no associated motivations, you’ll see an italic message.

You selected YES or UNSURE

The **Examples** block appears: here you must enter **at least 2 linguistic examples** illustrating the answer. For each example:

- **Example text:** the sentence in the language (e.g. *Mario ha letto il libro nuovo che gli ho consigliato*);
- **Transliteration:** transliteration if needed (e.g. for non-Latin alphabets);
- **Gloss:** morpheme-by-morpheme gloss (e.g. *book.M.SG.NOM*);
- **English Translation:** English translation;
- **Reference:** source (e.g. *Hauser, M. 1992. p. 12*).

Useful buttons:

- **+ Add another example:** adds a fresh empty card for a new example.
- **Remove:** deletes an example.
- **Copy:** copies the example to the **internal clipboard** (see below).
- **+ Import example from another answer...:** server-side search through examples already written for other questions/languages. Find the example (by text / translation / gloss), click it, and it gets **copied as a new example** in this question. **The imported example is an independent copy:** you can edit it freely without affecting the original.

⚠ **Important rule:** if you answer YES or UNSURE, you **must** provide **at least 2 non-empty examples**. If you try to save with fewer than 2, the system blocks the save with a warning and **highlights** the offending card in red. Same constraint applies to UNSURE.

Comments

At the bottom of each card there is a **Comments** field (free text) for unstructured notes/explanations/doubts.

Example clipboard

Handy utility to avoid retyping the same example in multiple questions of the same language:

1. In any example card, click **Copy** at the top-right → the example goes into the internal clipboard (a red dashed banner appears below your current examples).
2. Move to another question of the same language → you’ll see the same banner.

3. Click **Paste here** → the example is added as a **new** example in this question. The clipboard stays full so you can paste into more questions in a row.
4. **Clear** on the banner empties the clipboard.

⚠ The clipboard is **language-specific**. If you switch to a different language, the clipboard is automatically cleared to avoid carrying around an “orphan” example.

Saving a parameter block

At the bottom-right of the current block you see a **floating (sticky) toolbar** with two buttons:

Button	When to use it
Confident → Next (green)	Data is complete and verified. The parameter square turns green.
Unsure → Next (red)	You have doubts and want to save for later review. The parameter square stays yellow (flagged).

Both buttons **save all data of the current block** (answers + examples + motivations + comments + admin note if you're an admin) and then **automatically advance to the next parameter**.

If the save succeeds: no special feedback, you find yourself on the next parameter.

If the save fails, it can be one of these reasons:

- **400 missing examples**: you put YES or UNSURE on a question with fewer than 2 examples → an alert pops up and the offending card flashes red for a few seconds.
- **409 stale block**: meanwhile someone else (usually an admin) modified the same block. The system warns you and automatically reloads the page so you can review the other person's changes before retrying.
- **Network error**: generic alert, retry later.

Switching parameter without saving

If you've changed something and click another square without saving, the site **asks for confirmation** before discarding the current block's changes. Same if you close the tab or navigate away via breadcrumb: automatic protection.

Submit / Reopen / workflow

Once all the parameters you cared about are compiled, **and** the status is **Pending** or **Rejected** :

- **Submit for approval**: the language goes to **Waiting for approval**. The form **locks** and the admin receives the language in their dashboard.

When the status is **Rejected** :

- **Reopen:** the language goes back to `Pending` (after reading the rejection note). You can resume editing.

Remember: an admin can force the language status at any time and in any direction (see §9 of [documentation_en.md](#) for the full workflow).

What an admin sees in addition

Beyond the purple “Admin override” banner and the **Change Status** ▾ dropdown, an admin opening a language has:

- **Apply implicational condition(s)** (link at the top): opens the **Debug** page showing all init/final parameter values for the language, with the option to re-run the DAG just for this language.
 - **Admin notes** inside each parameter block: free-text note for (language, parameter). It is not exported and not visible to Users.
-

8. ADMIN ONLY — Parameters

Page: **Sidebar > Parameters** (`/admin/parameters`).

Parameter list

The PCM Hub

Dashboard

How to cite (edit)

DATA & TOOLS

Languages

Parameters

Parameters Graph

Questions

Motivations

Taxonomy

Table A

Filters

Accounts

History & Backups

Migration Import

Backup Restore

Instructions

Glossary

Dashboard / Parameters

MyAccountDarkLogout

Parameter Management

SEARCH

SCHEMA

TYPE

LEVEL

ACTIVE

Search every field...

All

All

All

All

108 of 108 parameters

Reset

Download parameters info (.pdf)

+ Full Parameters Backup

Add Parameter

ID	Name	Schema	Type	Level	#Q	#QS	Status	Actions
FGM	grammaticalized morphology	Availability	Subsetting	Eliminati (2026) ma non ancora disattivati	2	0	Active	PDFBackupEdit
FGA	grammaticalized agreement	Grammaticalization	Subsetting	Eliminati (2026) ma non ancora disattivati	1	1	Active	PDFBackupEdit
FGK	grammaticalized Case	Grammaticalization	Asymmetrical	Crisma et al (2025) - UPDATED	3	1	Active	PDFBackupEdit
SPK	grammaticalized (ultra)spatial Cases	Grammaticalization	Asymmetrical	Eliminati (2026) ma non ancora disattivati	1	1	Active	PDFBackupEdit
FGP	grammaticalized Person	Grammaticalization	Asymmetrical	Crisma et al (2025) - UPDATED	11	1	Active	PDFBackupEdit
FSP	semantic Person	Availability	Subsetting	Crisma et al (2025) - UPDATED	2	0	Active	PDFBackupEdit
FGN	grammaticalized Number	Grammaticalization	Asymmetrical	Crisma et al (2025)	6	1	Active	PDFBackupEdit

Table with: ID, name, position, schema, type, level of comparison, counters `questions_count` (regular questions) and `stop_count` (stop questions), active/inactive status.

At the top: search field, **Add parameter** button, **Export PDF (all)** (generates a PDF with all parameters).

Drag & drop to reorder parameters (`position` is saved automatically).

Edit parameter

The screenshot shows the 'Edit Parameter: FGM' page in The PCM Hub. The page is divided into a left sidebar, a main content area, and a right sidebar.

Left Sidebar: Contains navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters (highlighted), Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, and Glossary.

Main Content Area:

- Header:** 'Edit Parameter: FGM' with a 'Download PDF' button.
- Parameter ID:** FGM
- Position:** 1
- Name:** grammaticalized morphology
- Schema:** Availability
- Type:** Subsetting
- Level:** Eliminati (2026) ma non ancora dhr
- Short Description:** Distinguishes languages that have words containing bound morphemes for grammatical meanings (e.g., IE, Uralic, Semitic, Japanese) from languages that do not (e.g., Mandarin, Cantonese)
- Long Description:** test test2 aggiungo modifica
- Implicational Condition(s):** e.g. (+FGM | -ABC)
- Explanation of the Implicational Condition(s):** Textual explanation (optional)

Right Sidebar:

- Where Used:** Other parameters that depend on this one. (FGA: grammaticalized agreement)

Click the **Edit** button on a row → edit page with:

- ID (read-only, cannot be changed after creation);
- Name, position, schema, type, level of comparison;
- **Short description** (shown during compilation);
- **Long description** (shown in PDFs);
- **Implicational condition:** string with syntax like `+SPK & -DEM` (see §8.1);
- **Description of the implicational condition:** textual description;
- **Active:** active/inactive toggle (an inactive parameter is not offered for compilation and does not show in TableA);
- **Change note** (mandatory): why are you making this change. Goes to History.

To the left of the form you also see:

- the list of the parameter's **questions** with reorder drag & drop, and the option to open the question edit drawer directly from here;
- the history of **change notes** (ParameterChangeLog) with author and date.

Deactivating a parameter

Deactivate button: asks for confirmation + the **current admin's password** (safeguard). Once deactivated, the parameter is no longer offered for compilation and doesn't contribute to TableA / DAG / exports. Historical answers remain in the DB.

Implicational condition (notes)

The target parameter is “comparable” only if its `implicational_condition` is satisfied by the cited “ref” parameters.

- `+P` means: parameter `P` has value `+` for the language;
- `-P` : value `-` ;
- `0P` : value `0` (neutralized);
- `&` = AND, `|` = OR, `~` = NOT, `( )` = grouping.

Example: `+SPK & -DEM` means “parameter SPK is `+` AND parameter DEM is `-`”. If false, the target parameter is forced to `0` (with warning).

The condition is **syntactically validated** at save time: if malformed, the system rejects with an error message. If it cites non-existent or inactive parameters, it is **silently ignored** by the DAG (but the save goes through).

Parameters Graph (/admin/parameters/graph)

The screenshot displays the 'Parameters Graph' interface in PCM Hub. The sidebar on the left includes links to Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph (highlighted), Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, and Glossary. The main area shows the 'Parameters Graph' title, a breadcrumb 'Dashboard / Parameters / Graph', and user controls for 'MyAccount', 'Dark', and 'Logout'. Below the title is a subtitle: 'Implicational dependencies between parameters. Click a node to inspect its chain; double-click to open it for editing.' There are four filter dropdowns: Language (set to 'none'), Schema (set to 'All'), Type (set to 'All'), and Level (set to 'All'). A search bar with the placeholder 'Search ID or name...' and buttons for 'Fit all' and 'Reload' are also present. The graph itself is a complex network of nodes and edges. A legend indicates that clicking a node shows its 'CHAIN', with colors for 'selected' (yellow), 'antecedent' (green), and 'consequent' (orange). The right sidebar contains two panels: 'No node selected' with a message to click a node to see its dependency chain, and 'Implicational condition' which is currently empty.

Visualization of the **implicational DAG graph** in an interactive diagram. Useful to understand which parameters depend on which. Each node is a parameter, edges go from `ref` → `target`.

9. ADMIN ONLY — Questions

Page: **Sidebar > Questions** (`/admin/questions`).

Question list

The screenshot shows the PCM Hub interface for managing questions. The sidebar on the left contains navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions (highlighted), Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, and Glossary. The main content area is titled 'Questions' and includes a search bar with the placeholder 'Search every field (ID, parameter, text, instructions, template)...' and an 'Add Question' button. Below the search bar is a table of questions.

ID	Text Snippet	Type	Actions
FGM_Qa	Does the language have affixes or regular phonological alternations th...	Standard	Edit Delete
FGM_Qb	Does the language have roots which take different affixes/phonological...	Standard	Edit Delete
FGA_QSa	test Inactive	Stop	Edit Restore
FGA_Qa	Do you find alternations where a feature occurring on a word takes its...	Standard	Edit Delete
FGK_QSa	Are items designating the speaker, the addressee and the non-participa...	Stop	Edit Delete
FGK_Qa	Does the morphology of personal or relative pronouns occurring as argu...	Standard	Edit Delete
FGK_Qb	In nominal arguments, does the morphology of quantifiers, demonstrativ...	Standard	Edit Delete
FGK_Qc	In nominal arguments, does the morphology of nouns vary according to t...	Standard	Edit Delete
SPK_QSa	Is at least one of the simple spatial relations (stative location, dir...	Stop	Edit Delete
SPK_Qa	Does the language have morphological Case distinctions expressing spat...	Standard	Edit Delete
FGP_QSa	Do you find "indicative" (modus realis) clauses where the verb is inva...	Stop	Edit Delete
FGP_Qa	Do you find morphological alternations on the verb that depend on the ...	Standard	Edit Delete

All questions in the system (across all parameters). Filters by parameter, template_type, active/inactive, stop question. "Stop" questions are those that close a parameter's block.

Edit/Add question

To create a question, you must first choose its **parent parameter** (ID).

Main fields:

- **ID** (e.g. `FGM_01`, convention `<param_id>_<NN>`);
- **Text**: the question text shown to the linguist;
- **Template type**: classifies the question (useful for statistical filters);
- **Instruction** (always visible during compilation);
- **Instruction YES** (visible only if the answer is YES or UNSURE);
- **Instruction NO** (visible only if the answer is NO);
- **Example YES**: an illustrative example always visible to the linguist as reference;
- **Help info**: long text shown behind a "More info" button;
- **Is stop question**: tick if it closes the parameter;
- **Active**: tick to make it available for compilation;

- **Allowed motivations:** list of motivations selectable by the linguist when answering NO. Only the motivations ticked here appear in the compilation checkbox.
-

Destructive changes: the archive

⚠ Modifying a question can **invalidate** answers already given. Typical examples:

- changing `template_type` from `boolean` to something else;
- deleting the question;
- removing motivations from the “allowed” list.

In these cases, the system **does not lose** the data: it automatically moves the related

`Answer` / `Example` / `AnswerMotivation` to the **question archive** ([History](#) → [Old questions archive](#)). They remain readable read-only there, with a snapshot of the question as it was before the change.

10. ADMIN ONLY — Motivations

Page: **Sidebar > Motivations** (`/admin/motivations`).

The screenshot shows the PCM Hub Admin interface. The sidebar on the left contains the following links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, **Motivations** (highlighted), Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, and Glossary. The main content area is titled 'Motivations' and includes a search bar, an 'Add Motivation' button, and a table of predefined motivations.

Code	Description (Label)	Linked Questions	Actions
MOT001	The language has no personal/relative pronouns	FGK_Qa	Edit Delete
MOT002	There are no differences in the morphology of personal/relative pronouns depending on their syntactic function TEST TEST	FGA_Q5a FGA_Qa FGK_Qa FGH_Qb	Edit Delete
MOT003	The language has no Case morphology on nouns	DPQ_Qa	Edit Delete
MOT004	The language has Case morphology on nouns, but there are no morphological Case distinctions expressing spatial relations more complex than stative location, direction, and source	DPQ_Qa FGA_Q5a SPK_Qa	Edit Delete
MOT008	The language has no speech role designating items	FGP_Qf	Edit Delete
MOT009	Speech role designating items occur to the right of Adjectives/Numerals	FGP_Qf	Edit Delete
MOT010	The language has no pronominal Adjectives/Numerals	FGP_Qf	Edit Delete
MOT011	The language has articles	FGP_Qg	Edit Delete
MOT012	The order of cardinal numerals and possessives/adjectives meaning 'other', 'same/even' or 'unique'/the noun itself is fixed	FGP_Qg	Edit Delete
MOT014	Word order alternations of cardinal numerals and possessives/adjectives meaning 'other', 'same/even' or 'unique'/the noun itself are not associated with differences in the definite/non definite testinterpretation	FGP_Qg	Edit Delete

The **motivations** are the predefined reasons a linguist can tick when answering NO to a question. Example: “no corresponding structure”, “marginal phenomenon”, “phonetically reduced”.

List

Each motivation has:

- **Code:** unique identifier (e.g. `NO_STRUCT`);
- **Label:** the text shown to the linguist (can be long, multiline).

Edits

Standard CRUD. Motivations are never fully “deleted”: if used by some answer, their snapshot stays in the archive.

Motivations are **linked to questions** from the [Questions](#) page (field **Allowed motivations**), not from here.

11. ADMIN ONLY — Taxonomy

Page: **Sidebar > Taxonomy** (`/admin/taxonomy`).

The screenshot shows the PCM Hub Taxonomy admin interface. The sidebar on the left includes navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, Motivations, **Taxonomy** (selected), Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, and Glossary. The main content area is titled 'Taxonomy' and includes a breadcrumb 'Dashboard / Taxonomy'. It features a '+ New' button and a note: 'Click an item to view its children. Drag to reassign its parent. Renaming updates all linked languages.' Below this, there are three main sections: 'Top-Families' (listing Indo-European, Niger-Congo, etc.), 'Unassigned Subfamilies' (Romance, Celtic, etc.), and 'Unassigned Groups' (Italian, Spanish, etc.). Each section has a '+ New' button. The 'Top-Families' section lists various language families with their respective subfamily and language counts, and options to 'Edit' or 'Delete'.

Editor of the **language family hierarchy**:

```

Top Family (e.g. Indo-European)
├── Family / Subfamily (e.g. Romance)
│   └── Group (e.g. Italian-Romance)

```

Drag & drop across levels to reorder. Click a name to edit, "+" to add a child, trash to delete.

⚠ The three strings `top_level_family`, `family`, `grp` on Languages are **denormalized**: they're copied from the taxonomy at language save time. Editing the taxonomy **does not** automatically rename the strings on existing languages — you have to reopen and re-save the language to propagate the change.

12. ADMIN ONLY — Accounts

Page: **Sidebar > Accounts** (`/admin/accounts`).

Account list

Dashboard / Accounts MyAccount Dark Logout

Accounts

Search a user... Add Account

Administrators

Email	Name	Languages	Actions
admin@pcm.local	Admin PCM	All	Delete
admin@test.com	Admin	All	Delete

Users

Email	Name	Languages	Actions
user@test.com	user user	3 assigned	Assign Langs Delete

Public Users

Email	Name	Languages	Actions
-------	------	-----------	---------

Table with: email, name, surname, role (`admin/user/public`), number of assigned languages.

Buttons:

- **Add account** (`/admin/accounts/add`): create a new account.
- **Assign languages** on the row: opens a panel to select the language pool to assign to a user (multi-select).
- **Delete** on the row.

Add account

The screenshot shows the 'Add Account' form in the PCM Hub interface. The sidebar on the left lists various tools and data options, with 'Accounts' highlighted. The main content area is titled 'Create New Account' and includes the following fields and controls:

- Name** and **Surname**: Text input fields.
- Email**: Text input field.
- Temporary Password (min 8 characters)**: Text input field.
- Role**: A dropdown menu currently showing 'Standard User'.
- Buttons**: 'Cancel' and 'Create Account' buttons.

Form with: name, surname, email, temporary password (min 8 chars), **role** (dropdown: User / Admin / Public).

After creation, share the credentials with the user; on first login they should change the password.

⚠ Email validation: the system rejects with "Invalid email format" if the email is malformed (e.g. missing `.it`, double `@`, spaces). This prevents creating accounts that then can't log in.

Safeguards

- **You cannot delete yourself** (have another admin do it).
- **You cannot delete the last admin** (the site would be left without an administrator).

Reassigning languages

The **Assign languages** action **replaces** the current pool: if a user had 3 languages assigned and you tick only the other 2 keeping them in the new 5, the original 3 go back to "unassigned" (assignable to another user).

13. ADMIN ONLY — Table A

Page: **Sidebar > Table A** (/tablea).

The PCM Hub

Dashboard

How to cite (edit)

DATA & TOOLS

Languages

Parameters

Parameters Graph

Questions

Motivations

Taxonomy

Table A

Filters

Accounts

History & Backups

Migration Import

Backup Restore

Instructions

Glossary

Dashboard / Table A

MyAccountDarkLogout

Table A

Parameters ViewQuestions View

LANGUAGE FILTERS

SPECIFIC LANGUAGES

☐ Afrikaans (Afk)

☐ Aidone (Aid)

☐ Arabic (Ar)

☐ Archi (Arc)

☐ ...

TOP FAMILY

All

SUBFAMILY

All

GROUP

All

HISTORICAL

Both

PARAMETERS FILTERS

SCHEMA

All

TYPE

All

LEVEL

All

Apply Filters

Reset

Download Data

Select AllDeselect All

☐	ID	Parameter Name	Implicational conditions	Sic	NCa	Sp	Fr	Ptg	Rm	CGB	Grk	CyG	E	Du	Dz
☐	FGM	grammaticalized morphology		+	?	?	?	?	?	?	?	?	?	?	?
☐	FGA	grammaticalized agreement	+FGM	-	?	?	?	?	?	?	?	?	?	?	?

TableA is the project’s flagship matrix: **one language per column, one parameter (or question) per row**, with the consolidated value in each cell (+ , - , 0 , ?).

Modes

- **Params** (default): rows = parameters.
- **Questions**: rows = individual questions.

Toggle at the top.

Filters

- **Languages**: top family / family / group / historical / specific (per-language checkboxes in the lower section);
- **Parameters**: schema, type, level of comparison;
- **Questions**: template, stop question (visible only in Questions mode).

After choosing filters, click **Apply** (or equivalent) to recompute the matrix.

Manual selection

Beyond filters, **each table row** has a checkbox: lets you further restrict the analysis to a subset of parameters/questions.

34

Exports and analyses

Download / Analysis ▾ dropdown at the top:

- **XLSX (parameters)** — Excel with citation header.
- **CSV (transposed)** — standard analytical format.
- **Distance matrices (.txt)** — Hamming and Jaccard matrices.
- **Geographic distances (.zip)** — GCD km matrix.
- **Dendrograms (.png)** — UPGMA dendrograms.
- **Cluster map (.html)** — interactive HTML map of geographic clusters.
- **PCA (.png)** — principal components analysis.
- **Mantel test (.zip)** — modal with three checkboxes (GCD, Hamming, Jaccard) to choose which matrices to correlate. Output: correlation tables + plots.

⚠ Distances and clusters are meaningful only on sufficiently large and homogeneous sets. Languages without coordinates are excluded from geographic distances and the export's `X-Skipped-Languages` header lists them.

14. ADMIN ONLY — Filters (Queries Q1–Q10)

Page: **Sidebar > Filters** (/queries).

The screenshot shows the PCM Hub interface. On the left is a sidebar titled 'The PCM Hub' with a menu including 'Dashboard', 'How to cite (edit)', 'DATA & TOOLS', 'Languages', 'Parameters', 'Parameters Graph', 'Questions', 'Motivations', 'Taxonomy', 'Table A', 'Filters' (highlighted in red), 'Accounts', 'History & Backups', 'Migration Import', 'Backup Restore', 'Instructions', and 'Glossary'. The main area is titled 'Dashboard / Filters' and 'Filters & Queries'. It features a 'Queries Configuration' panel on the left with 10 predefined queries (Q1-Q10) and a search bar on the right with a 'PARAMETER' dropdown and a 'Search' button.

10 predefined views to query the database in a focused way. Menu on the left (collapsible), main area on the right.

Query	Question
Q1	Show implicational condition(s) (per parameter)
Q2	Show parameter values for all languages (per parameter)
Q3	Show why a parameter is neutralized (per language) ★
Q4	Parameters with value + (per language)
Q5	Parameters with value − (per language)
Q6	Parameters with value 0 (per language)
Q7	Comparable parameters (per pair of languages)
Q8	Questions with answer YES (per language)
Q9	Questions with answer NO (per language)
Q10	Show answers and examples (per question)

Q3 — the richest

It's the **DAG debug view**: pick a language and a neutralized parameter, and you get:

- the parameter's implicational condition;
- for each `ref` cited in the condition, what its value is (`value_eval` if present, else `value_orig`);
- whether the condition is satisfied or not;
- the list of linguist Answers that contributed to each ref's `value_orig` .

Useful to understand **why** a particular parameter is not being compared in a particular language.

Q10 — answers and examples

Q10 has 2 optional filters (language + parameter) to **restrict the dropdown of available questions**. Leaving them empty, the dropdown shows all the ~N hundreds of system questions.

15. ADMIN ONLY — History, Backups, Migration, Backup Restore, Import Excel, Recompute

Everything related to **audit, snapshots, mass import/export**.

History & Backups (/admin/history)

The site's **audit hub**. The page has **4 tabs** at the top: each one answers a different question ("who changed what", "who answered what", "do I have a complete snapshot somewhere", "what happened to the old version of a question?"). The tabs share the same filter-table layout but populate different data.

"Change history" tab

The screenshot shows the PCM Hub interface. The sidebar on the left contains navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups (highlighted), Migration Import, Backup Restore, Instructions, and Glossary. The main content area is titled 'History & Backups' and has four tabs: Change history (selected), Answer changes, Full backups (languages & parameters), and Old questions archive. Below the tabs are filters for ENTITY TYPE (All), ENTITY ID (e.g. FGM, FGM_01...), OPERATION (All), SOURCE (All), USER (All), and FROM (DAY) (dd/mm/yyyy). There is also a TO (INCLUSIVE) filter and a TEXT SEARCH field. A 'Reset' button and an 'Apply filters' button are present. Below the filters, it says '670 total versions'. A table displays the history of changes with columns: When, What, Op., Source, Author, Note, and Detail. The table shows several rows of changes, including updates to parameters and motivations.

When	What	Op.	Source	Author	Note	Detail
06/05/2026, 14:46:20	Parameter PKP — PKE with all persons	Updated	Manual	Admin	Reactivated	Open
06/05/2026, 14:45:55	Parameter PKP — PKE with all persons	Updated	Manual	Admin	Deactivated: test	Open
06/05/2026, 14:41:07	Parameter FGM — grammaticalized morphology	Updated	Manual	Admin	modificato	Open
06/05/2026, 14:37:11	Motivation 10 — Word order alternations of cardinal numerals and possessives	Updated	Manual	Admin	—	Open
02/05/2026, 20:08:56	Parameter FGM — grammaticalized morphology	Updated	Manual	Admin	Test edit. rimossa explanation of implicational condition	Open
02/05/2026, 20:08:23	Parameter	Updated	Manual	Admin	Test edit. bla vla	Open

Table of **all changes** to *non-Answer* entities in the system: **parameter**, **question**, **motivation**, **language**. Designed to answer "who modified parameter FGM and why", "when was motivation MOT042 created?", "who deleted language XYZ last month?".

Available filters: **Entity type** (dropdown), **Entity ID** (e.g. **FGM**, **FGM_01**), **Operation** (create / update / delete), **Source** (manual = panel edit, excel_import = arrived from an Excel import, system = generated by the app), **User** (who did it), **From / To** (date range), **Text search** (searches the ID or note). The **Reset** button clears all filters, **Apply filters** re-runs the query.

Click **Open** on a row (or the row itself) → a **side drawer** opens with the **before/after diff**: for each modified field, the old value in red on the left and the new in green on the right, plus a collapsible full snapshot at the bottom.

“Answer changes” tab

The PCM Hub

Dashboard

How to cite (edit)

DATA & TOOLS

Languages

Parameters

Parameters Graph

Questions

Motivations

Taxonomy

Table A

Filters

Accounts

History & Backups

Migration Import

Backup Restore

Instructions

Glossary

Dashboard / History & Backups

MyAccountDarkLogout

History & Backups

Change historyAnswer changesFull backups (languages & parameters)Old questions archive

ENTITY ID

OPERATION

SOURCE

USER

FROM (DAY)

TO (INCLUSIVE)

TEXT SEARCH

e.g. FGM, FGM_01...

All

All

All

dd/mm/yyyy

dd/mm/yyyy

Search ID and note...

13530 total versions

ResetApply filters

When	What	Op.	Source	Author	Note	Detail
07/05/2026, 09:37:15	Answer cB:FGM_Qb	Created	Manual	Admin	—	Open
02/05/2026, 19:25:21	Answer Sic:FGA_Qa	Created	Manual	user user	—	Open
02/05/2026, 19:23:55	Answer Sic:FGM_Qb	Updated	Manual	user user	—	Open
02/05/2026, 19:23:43	Answer Sic:FGM_Qa	Created	Manual	user user	—	Open
02/05/2026, 19:23:43	Answer Sic:FGM_Qb	Created	Manual	user user	—	Open
01/05/2026, 22:08:46	Answer Afk:FGK_Qb	Updated	Manual	Admin	—	Open
01/05/2026, 22:08:46	Answer FGK_Qb	Updated	Manual	Admin	—	Open

The “Answer-focused” version of the same table. Every time a linguist (or admin) clicks **Confident** → **Next** or **Unsure** → **Next** in a parameter block, a row appears here for every touched answer. Designed to answer “when did user X put YES on question FGM_Qb for Italian?” or “who changed the French answer for parameter SPK on day Y?”.

Same experience as the other tab: filters, table, click a row → drawer with diff. The split exists only because Answers change much more frequently than other entities (every block save) and mixing them with parameters/questions would make the “Change history” tab unreadable.

“Full backups (languages & parameters)” tab

The screenshot shows the PCM Hub interface. The left sidebar contains navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups (active), Migration Import, Backup Restore, Instructions, and Glossary. The main content area is titled 'History & Backups' and includes tabs for 'Change history', 'Answer changes', 'Full backups (languages & parameters)', and 'Old questions archive'. The 'Full backups (languages & parameters)' tab is active, showing sub-tabs for 'Languages' and 'Parameters'. A card titled 'Create global languages backup' provides instructions and a form to create a snapshot of all languages with an optional note. A table below shows no existing backups.

List of **complete snapshots** of the system, grouped by timestamp. Sub-tabs **Languages** or **Parameters** (toggle at the top), selecting two different snapshot types:

- **Language backups** are created from **Languages** → **+ Full Languages Backup** and contain all languages with answers and examples at the moment of capture.
- **Parameter backups** are created from **Parameters** → **+ Full Parameters Backup** and contain the state of all parameters (definitions, implicational conditions, descriptions) at capture time.

You can also **create one on the fly** from here: the top card has an optional **Note** field and a **+ Create languages backup now** (or **+ Create parameters backup now** on the other sub-tab) button. Clicking a row in the table opens the backup detail, with the option to download an Excel of all data for audit or external restore.

“Old questions archive” tab

The screenshot shows the PCM Hub interface. On the left is a sidebar with 'The PCM Hub' logo and a list of navigation items: Dashboard, How to cite (edit), DATA & TOOLS (Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, Migration Import, Backup Restore, Instructions, Glossary). The 'History & Backups' item is highlighted in red. The top bar shows 'Dashboard / History & Backups', 'MyAccount', 'Dark' theme toggle, and 'Logout'. The main content area is titled 'History & Backups' and has four tabs: 'Change history', 'Answer changes', 'Full backups (languages & parameters)', and 'Old questions archive' (which is active). Below the tabs is a text box explaining that each row groups archived versions of a single question, created when a question is saved with the 'Save and delete the linked data' button. A search bar is provided with the placeholder 'Search by question ID, parameter, archived text or note...'. Below the search bar is a table with the following data:

Question ID	Parameter	Archived versions	Latest archive	Actions
FGM_Qa	FGM grammaticalized morphology	1	30/04/2026, 10:11:55 by Admin	Show versions

When a question is modified destructively (see §9 — `template_type` change, deletion, removal of motivations) or deleted, its Answers/Examples/AnswerMotivations are **not lost**: they are moved here to the archive together with a snapshot of the question text as it was.

Each row in the table groups **all archived versions of the same question** (e.g. if FGM_Qa was “broken” 3 times, you see 3 versions here). The **Show versions** button opens the detail: for each version you see the original text, the motivations allowed at the time, and all historical answers from every language to that version of the question.

Useful when you have to demonstrate to a reviewer “the original linguistic data still exists, even if the question has since been modified”.

Migration Import (/admin/migration-import) — dangerous

The screenshot shows the PCM Hub interface. The sidebar on the left contains navigation links: Dashboard, How to cite (edit), DATA & TOOLS, Languages, Parameters, Parameters Graph, Questions, Motivations, Taxonomy, Table A, Filters, Accounts, History & Backups, **Migration Import** (highlighted in red), Backup Restore, Instructions, and Glossary. The main content area is titled 'Restore Database' and includes a warning: 'Operazione distruttiva'. Below the warning, there is a list of actions: 'Con wipe attivo, vengono cancellate tutte le tabelle dati', 'Gli utenti esistenti NON vengono toccati', 'Tutte le risposte importate sono salvate come pending', and 'Pensato per essere usato una sola volta'. A section titled 'Contenuto atteso del bundle' lists the files in the migration bundle and their corresponding data structures.

Operazione distruttiva

- Con wipe attivo, vengono **cancellate tutte le tabelle dati** (lingue, parametri, domande, risposte, esempi, motivazioni, glossario, snapshots, change logs, tassonomia).
- Gli utenti esistenti NON vengono toccati. Viene comunque (ri)creato l'admin di default con le credenziali da env `ADMIN_EMAIL/ADMIN_PASSWORD` (default: `admin@pcm.local/admin`).
- Tutte le risposte importate sono salvate come **pending** e il DAG viene eseguito per ogni lingua a fine import (Tabella A e dashboard saranno subito popolate).
- Pensato per essere usato **una sola volta** al primo avvio del nuovo sito. Dopo, nascondi/rimuovi questo bottone.

Contenuto atteso del bundle

```
PCM_migration_<ts>.zip
├── 00_languages.xlsx          (anagrafica + family/top/group)
├── 01_motivations.xlsx        (ID, Code, Label)
├── 02_parameters.xlsx         (ID, Position, Name, ...)
├── 03_questions.xlsx          (ID, Parameter ID, Text, ...)
├── 04_question_allowed_motivations.xlsx (Question ID, Motivation Code)
├── 06_glossary.xlsx           (Word, Description) [opzionale]
├── 08_unsure_flags.xlsx       (Language ID, Parameter ID) [opzionale]
├── data/
│   └── <Lingual>.xlsx        (foglio Database_model)
└── ...
```

⚠ Item shown in **red** in the sidebar because it can **wipe the database**.

Used to import the **legacy site's ZIP bundle** (Django legacy) during go-live. Typical procedure (see `DEPLOY_PROCEDURA.txt`):

1. Download the bundle from the legacy site.
2. Go to Migration Import.
3. Select the ZIP file.
4. Tick **wipe = true** to clean the DB before importing.
5. Click **Import**.
6. Wait (can take minutes, there are progress indicators).

After import the admin password is **reset to** `ADMIN_PASSWORD` from the Portainer env vars (change the password **after** the import, not before).

Validation: 200MB cap on the ZIP, anti-zip-bomb, path-traversal checks on file names.

Backup Restore (/admin/backup-restore)

The PCM Hub

Dashboard

How to cite (edit)

DATA & TOOLS

Languages

Parameters

Parameters Graph

Questions

Motivations

Taxonomy

Table A

Filters

Accounts

History & Backups

Migration Import

Backup Restore

Instructions

Glossary

Dashboard / Backup Restore

MyAccountDarkLogout

Backup Restore

Carica un file **PCM_backup_*.zip** generato via *Languages* → *Download data* → *Export backup (.zip)* per ripristinare schema, metadati lingue, glossario e dati di compilazione.

Bundle accettato

- schema.xlsx — Motivations / Parameters / Questions / QAM
- languages_metadata.xlsx — metadati di tutte le lingue
- glossary.xlsx — Word / Description
- languages/<ID>.xlsx — un file per lingua con risposte, esempi, motivazioni e admin notes

Backup ZIP file

Choose FileNo file chosen

☐ **Wipe data tables before restore** (destructive)

Start restoreNeed Migration Import instead?

Restore of a **ZIP bundle exported** from *Languages* → *Download Data* → *Export backup (.zip)*. Lets you restore a specific set of languages from a previous snapshot, without wiping the DB.

Import from Excel (/admin/import-excel)

The PCM Hub

Dashboard / Import Excel

MyAccount Dark Logout

Import from Excel

Upload an Excel file to import schema (Motivations, Parameters, Questions, QAM) and/or filled-in data of a single language (sheet Database_model1).

Strategy

- **Schema** (Motivations / Parameters / Questions / QAM): *strict update*. If the `id` exists in the DB, the fields are replaced with the value from the file. If the `id` does not exist, the row is an error (skipped, included in the report). Entities not mentioned in the file remain untouched.
- **Database_model**: *full replace* of the language. All answers/examples/motivations for the language are deleted, then only valid rows are inserted. Invalid rows remain empty (unanswered) and are included in the report.
- **Cascading errors**: if a motivation/question fails, the dependent rows generate an explicit error (never silent).

File Excel (.xlsx, max 50 MB)

Choose File No file chosen

Start Import Cancel

Structured Excel import. 50 MB cap on the file. The sheet must follow the round-trip format ([documentation_en.md §10.4](#)).

Recompute final values

⚠ Reached **from Languages** → **Recompute final values**, no dedicated page.

Re-runs the **implicational DAG** for **all** languages in the system. Use:

- after important changes to implicational conditions;
- after a bulk import;
- if you suspect some `value_eval` is stale.

It's an async job (can take minutes on large datasets). Toast at the bottom-right shows progress. When done: "Recompute complete" message, or "Recompute completed with N error(s) over M language(s). See server logs for details" if some language has problems.

16. Common errors and what to do

“Form locked by the current language status”

The language status is `Waiting for approval` or `Approved`, and you are not an admin. Wait for the decision, or (if you are the assigned user and the language is `Rejected`) click **Reopen**.

“Question X needs at least 2 valid examples when answering YES or UNSURE”

You picked YES or UNSURE on a question with fewer than 2 non-empty examples. Add/complete the examples and retry.

“This block has been modified by another session...”

Another user (usually an admin) modified the same block while you were working on it. The page automatically reloads: review the other person’s changes before retrying your save.

“Wrong email or password”

Wrong credentials. If you don’t remember: contact an admin (email reset is not yet active).

“Too Many Requests” at login

You typed the wrong password 5 times within a minute. Wait 1 minute and retry.

“Invalid email format”

You’re creating an account or changing email with an invalid value (e.g. missing `.it`, space, double `@`). Fix it.

“You cannot delete your own account”

Have another admin do it.

“You cannot delete the last remaining administrator”

Create a second admin **before** deleting the existing one.

The “Submit for approval” button is missing

- Language status is already `Waiting for approval` or `Approved`: nothing to do.
- Status is `Rejected`: see **Reopen**, not Submit.
- You’re not the assigned user for this language: only the assigned user can Submit.
- You’re an admin: admins don’t submit (they go directly through approve/reject).

The site logged me out unexpectedly

Token expired (30 min of inactivity). Log in again.

I changed the taxonomy but languages still show the old names

The `top_family` / `family` / `group` strings on languages are denormalized. Reopen each affected language and Save (even without changing anything) to propagate the new names. To do this in bulk: use Excel export/import.

17. Quick interface glossary

Term	Meaning
Wizard / squares	the row of small buttons at the top of the compilation page, one per parameter
Block / parameter block	the set of all questions of a parameter for one language
Confident / Unsure	the two save buttons of the block. Confident = everything verified; Unsure = I have doubts, will recheck
Stale block (409)	a block that was modified by another session while you were editing
Submit for approval	sending the language to the admin for review
Reopen	bringing a Rejected language back to Pending
Approved / Rejected / Pending / Waiting	the 4 states of a language
Admin override	the purple banner reminding the admin they're editing a locked language
Force status	the admin dropdown to override a language's status bypassing the workflow
Implicational DAG	the parameter dependency graph (+SPK & -DEM) deciding whether a value should be neutralized
value_orig vs value_eval	orig = aggregated from the linguist's answers; eval = orig post-DAG (can be 0 if neutralized)
Comparable	a parameter is "comparable" between two languages if both have a non-null value_eval
Implicational condition	the string like +SPK & -DEM that triggers neutralization
Neutralized	a parameter forced to 0 because its implicational condition is not satisfied
Flagged / Unsure parameter	parameter marked "uncertain" by the linguist (yellow square)
Red parameter	in the Admin Dashboard, "red" parameters = unsure or incomplete
Hamming / Jaccard distance	distance metrics between languages based on parameter values
GCD	Geographic Distance, kilometric distance computed from lat/lng
Mantel test	statistical test measuring correlation between distance matrices
Top family / Family / Group	the 3 levels of the linguistic taxonomy
Glossary	dictionary of technical terms

How to cite	the page with the PCM Hub bibliographic citations
ZIP bundle	the compressed archive used for migration / backup-restore
Recompute	the job that re-runs the DAG over all languages

For doubts, suggestions or bug reports: write to pcm_Lab@unimore.it (link at the bottom of every page on the site).